SMART GLOVE FOR REAL-TIME WOUND AND INFECTION DETECTION

Submitted in partial fulfilment of the requirements
for the award of the degree in

BACHELOR OF TECHNOLOGY

in

BIOMEDICAL ENGINEERING

by

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DEPARTMENT OF BIOMEDICAL ENGINEERING

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DEEMED TO BE UNIVERSITY
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APRIL 2026

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BONAFIDE CERTIFICATE

This is to certify that this project report is the bonafide work of

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LIST OF ABBREVIATIONS

ABBREVIATIONS	DEFINITION
ESP32	Espressif 32-bit Microcontroller
ESP32-CAM	ESP32 Camera Module with Wi-Fi
IR	Infrared
RGB	Red, Green, Blue
I²C	Inter-Integrated Circuit
VCC	Voltage Common Collector (Power Supply Pin)
GND	Ground
GPIO	General Purpose Input/Output
MLX90614	Infrared Temperature Sensor Module
DHT11	Digital Humidity and Temperature Sensor
TCS34725	RGB Color Sensor Module
LED	Light Emitting Diode
USB	Universal Serial Bus
°C	Degree Celsius
IDE	Integrated Development Environment
AI	Artificial Intelligence
IoT	Internet of Things

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ABSTRACT

This project presents a smart wearable system for real-time wound and infection detection in dogs. The system integrates ESP32 DevKit V1 and ESP32-CAM with multiple sensors, including an MLX90614 infrared temperature sensor, DHT11 humidity sensor, and a TCS34725 colour sensor to monitor wound conditions. An RGB LED indicator is also incorporated to provide instant visual alerts for different health states such as normal, warning, and infection conditions. An FTDI programmer is used for programming and uploading code to the ESP32-CAM module. The ESP32-CAM captures images of the wound area, which are stored and analysed using a pre-trained Convolutional Neural Network (CNN) model to classify different skin conditions such as healthy, inflammatory infection, ringworm, and demodicosis. In addition to image-based analysis, sensor data is collected and processed using noise removal and threshold techniques to improve accuracy and reliability. The system enables real-time monitoring and displays the results through a mobile application and cloud platform. The RGB LED provides immediate on-device status indication, making the system more user-friendly even without a mobile or internet connection. This non-invasive and cost-effective solution supports early detection of skin diseases in dogs, enabling timely treatment and improved health outcomes.

Keywords: Smart Wearable System, Wound and Infection Detection, ESP32-CAM, Convolutional Neural Network (CNN), Sensor-based Monitoring

MAJOR DESIGN CONSTRAINTS AND DESIGN STANDARDS TABLE

Student Group	DHIVYABHARATHI S (221241101015)	GIRIJJAA S N (221241101021)	HARIKARAN S (221241101026)
Project Title	Smart Glove for Real-Time Wound and Infection Detection		
Program Concentration Area	Biomedical Instrumentation, Embedded Systems		
Constraints Examples	Sensor integration, real-time data processing, wearable ergonomics		
Economic	YES		
Environmental	YES		
Sustainability	YES		
Implementable	YES		
Ethical	Followed professional ethics in animal care and data handling		
Health and Safety	YES		
Social	Improves access to early veterinary diagnostics in underserved areas		
Political	NA		
Others	Real time infection detection using temperature, humidity, and color sensors		
Standards			
1.	IEEE 21451 – Smart Transducer Interface Standard		
2.	ISO 13485 – Medical Device Quality Management		
Prerequisite Courses for the Major Design Experience	1. Biomedical Instrumentation 2. Embedded System Design 3. Sensors and Signal Processing		

CHAPTER 1

INTRODUCTION

Advancements in technology have significantly improved healthcare systems, creating opportunities to enhance animal care through intelligent and automated solutions. Monitoring the health of animals, particularly detecting wounds and infections at an early stage, remains a critical challenge due to the lack of effective communication from animals and the limitations of manual observation methods. In recent years, the integration of Internet of Things (IoT), cloud computing, and machine learning has enabled the development of smart and real-time monitoring systems that can assist in accurate diagnosis and continuous health assessment. This chapter presents the foundation of the proposed Smart Glove system for wound detection and monitoring in dogs. It outlines the background of the study, identifies the existing challenges, and emphasizes the need for an efficient and reliable system. Further, it describes the proposed solution, technologies used, system overview, and its practical applications in animal healthcare.

1.1 BACKGROUND OF THE STUDY

Animal healthcare is an important aspect of maintaining the health and well-being of pets, especially dogs, which are highly prone to wounds and skin infections due to their active nature and exposure to different environmental conditions. These wounds may occur due to injuries, bites, or infections and can become severe if not identified and treated at an early stage. Early detection plays a key role in preventing complications and ensuring proper treatment and faster recovery. However, animals cannot clearly express their pain or discomfort, making it difficult for pet owners to recognize health issues in the initial stages.

At present, wound detection is mainly carried out through manual observation by veterinarians or pet owners. This method is not always reliable, as it depends on human judgment and may fail to identify minor or hidden infections. In addition, continuous monitoring of the animal's condition is not always possible, especially in busy or remote environments. These limitations may lead to delayed diagnosis and worsening of the condition.

Moreover, factors such as temperature, humidity, and changes in skin color, which are important indicators of infection, are often not monitored systematically. This lack of objective data makes accurate assessment more challenging. Therefore, there is a growing need for advanced technological solutions to support early detection and continuous monitoring in animal healthcare.

1.2 PROBLEM STATEMENT

Wound detection and infection monitoring in animals, especially dogs, is a challenging task in current veterinary practices. Most existing methods rely on manual observation by veterinarians or pet owners, which may not always be accurate or reliable. Early-stage wounds and infections are difficult to identify, as animals cannot clearly express their discomfort. This often leads to delayed diagnosis and improper treatment. In addition, continuous monitoring of the wound condition is not possible using traditional methods.

Environmental factors such as humidity and temperature, which play a significant role in wound healing and infection spread, are not regularly monitored. This lack of real-time data makes it difficult to assess the severity of the wound accurately. Moreover, existing systems do not provide an integrated solution that combines image-based analysis and sensor data for effective wound classification. There is no single system that captures wound images, collects sensor data, and performs analysis.

1.3 NEED FOR THE SYSTEM

The need for an efficient system for detecting wounds and infections in animals has increased due to the limitations of traditional methods. In most cases, wound detection is carried out through manual observation, which may not always provide accurate results and can lead to delayed diagnosis. Since animals are unable to express their discomfort clearly, early identification of infections becomes difficult, increasing the risk of severe health complications.

Additionally, continuous monitoring of wound conditions is not feasible using conventional approaches. Important factors such as temperature, humidity, and color changes, which influence wound healing, are not consistently measured. This creates a gap in proper assessment and timely treatment.

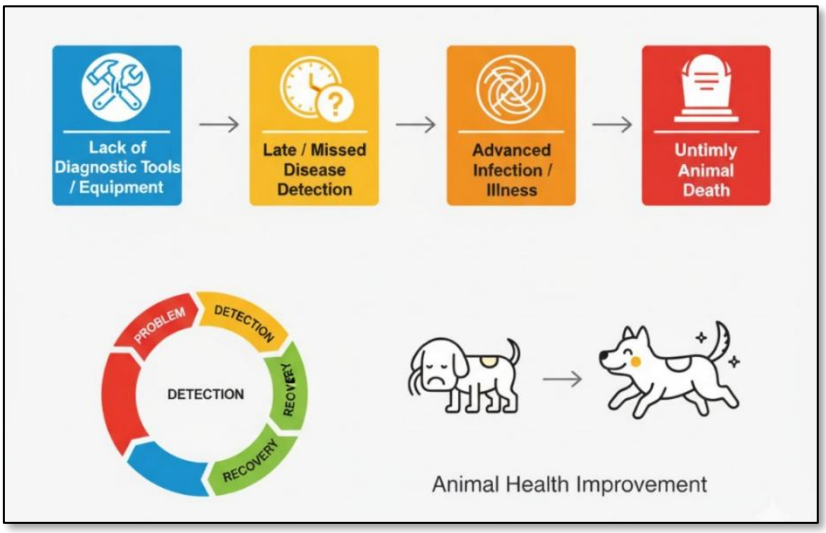

Fig. 1.1 Need for Early Detection

Therefore, there is a need for an intelligent and automated system that can provide real-time monitoring and accurate analysis of wound conditions. Such a system can assist in early diagnosis, improve treatment decisions, and enhance overall animal healthcare management.

1.4 PROPOSED SYSTEM

The proposed system is a Smart Glove designed for real-time detection of wounds and infections in dogs using IoT and machine learning techniques. The system uses an ESP32-CAM module to capture images of the wound and an ESP32 DevKit V1 to collect sensor data such as temperature, humidity, and color values. The collected data is processed using basic preprocessing techniques like noise removal and thresholding to improve accuracy. The processed wound image along with sensor data is then given to a trained machine learning model, which classifies the wound into different categories based on the condition.

The data and results are managed using the cloud platform ThingSpeak. A mobile application developed using MIT App Inventor is used to display the results, including wound image, sensor readings, and classification output in real time. Additionally, an RGB LED is used to indicate the severity of the infection. The proposed system provides a simple, portable, and effective solution for early detection and monitoring of animal wounds. The system also enables continuous monitoring and remote access to data, improving convenience for both veterinarians and pet owners. This integration of IoT and intelligent analysis enhances timely decision-making and overall animal healthcare management.

1.5 TECHNOLOGIES USED

The proposed system utilizes a combination of modern technologies to achieve efficient wound detection and monitoring. Internet of Things (IoT) technology is used for real-time data collection from sensors and communication between hardware components. Machine learning techniques are applied to analyze the wound images and sensor data for accurate classification of different infection types.

The system uses the cloud platform ThingSpeak to store and manage the collected data, including sensor readings, wound images, and classification results. This enables easy access and monitoring of data in real time. A mobile application developed using MIT App Inventor is used to display the output to the user. It provides a simple interface to view the wound condition, sensor values, and analysis results. These technologies work together to provide a reliable, real-time, and user-friendly system for animal wound detection and monitoring.

1.6 SYSTEM OVERVIEW

The proposed system is a smart glove developed to detect and monitor wounds and infections in dogs in real time. It combines image processing, sensor data acquisition, and machine learning techniques to provide accurate results. The system consists of two main hardware components: the ESP32-CAM for capturing wound images and the ESP32 DevKit V1 for collecting sensor data such as temperature, humidity, and color values. The captured image and sensor data are preprocessed using techniques like noise removal and thresholding to improve the quality and accuracy of the data.

This processed data is then given as input to a trained machine learning model, which analyzes the wound condition and classifies it into different categories. The data, including sensor readings, processed images, and classification results, are stored in the cloud platform ThingSpeak. A mobile application developed using MIT App Inventor retrieves this information and displays it to the user in real time for easy monitoring. Additionally, an RGB LED is used to provide immediate visual indication of the infection level. Overall, the system offers a simple, portable, and efficient solution for early detection and continuous monitoring of animal wounds.

1.7 APPLICATION

The proposed smart glove system has various applications in the field of animal healthcare. It can be effectively used by veterinarians for early detection and accurate diagnosis of wounds and infections in dogs. The system helps in reducing manual effort and provides real-time data, which improves the quality of treatment.

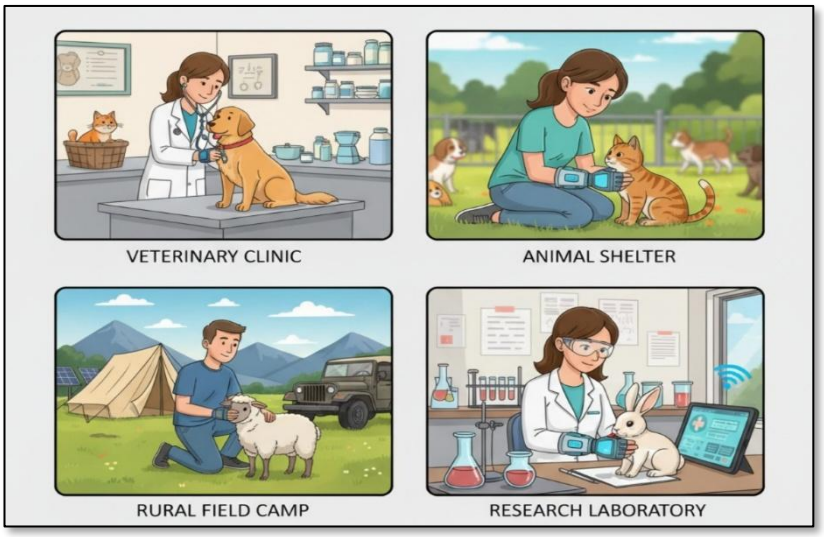

Fig. 1.2 Practical Applications

Pet owners can also use this system to monitor the health condition of their pets at home without the need for frequent visits to veterinary clinics. This is especially useful for early identification of infections and timely medical attention. In addition, the system can be used in animal shelters and pet care centers where continuous monitoring of multiple animals is required. It provides a simple and efficient way to track wound conditions and prevent the spread of infections. Overall, the system is useful in improving animal healthcare management by enabling early detection, continuous monitoring, and timely treatment of wounds and infections.

This chapter has outlined the key aspects of the proposed Smart Glove system, including its background, need, methodology, and applications in animal healthcare. It highlights the importance of early detection and continuous monitoring of wounds in dogs. The next chapter presents a review of existing research in AI-based wound detection, sensor integration, and smart veterinary systems, providing insights and identifying gaps that support the proposed work.

CHAPTER 2

LITERATURE REVIEW

This chapter reviews key research in AI-based wound detection, sensor integration, and smart veterinary diagnostics. It highlights how technologies like CNNs, YOLO models, infrared thermography, and IoT systems have been applied to monitor animal health. These studies lay the technical foundation for developing a smart glove capable of real-time noninvasive infection detection using temperature, humidity, and colour sensors, particularly valuable in rural and field settings where access to advanced diagnostic tools is limited. The review also identifies gaps in existing systems, such as limited species generalization and high equipment costs. These insights guide the design of a more accessible and scalable veterinary solution.

‘’Serena K., Patel D., Wong L. (2019): Convolutional Neural Networks for Wound Detection” This study explores the use of Convolutional Neural Networks (CNNs) to automate wound detection and classification using medical image datasets. It demonstrates how AI-based image processing can enhance diagnostic accuracy and reduce manual effort. Although focused on human wounds, the methodology offers a strong foundation for veterinary adaptation. The paper highlights AI’s role in clinical imaging but is paywalled and lacks animal-specific validation, making direct application to veterinary settings a challenge.
Additionally, the study emphasizes feature extraction techniques that improve classification performance. However, it does not consider real-time implementation or integration with sensor-based data.

“Chen Y., Kumar R., Singh A. (2022): Dog Skin Disease Recognition Using Image Segmentation and CNN” This paper applies CNN-based image segmentation to classify canine skin diseases such as ringworm and scabies. GPU acceleration improves training speed and recognition accuracy. The study demonstrates AI’s potential in veterinary diagnostics and supports mobile deployment. However, it is limited to a narrow disease set and small dataset size, which restricts generalization across species and broader conditions. It serves as a useful reference for targeted skin disease detection. The model also shows good performance in identifying distinct visual patterns in skin conditions.

“Smith J., Ar ora P., Lin Q. (2024): AI-Driven Veterinary Diagnostics” This paper investigates the integration of artificial intelligence into mobile applications for veterinary diagnostics. It enables real-time disease detection and remote monitoring through cloud-based platforms. The study discusses data pipelines and mobile health frameworks but lacks specific focus on wound detection. While promising for general veterinary care, its broad scope and paywall access limit its direct relevance to wound- specific applications in field settings. The study also highlights the importance of cloud-based data synchronization for continuous monitoring.

“Hossam M., El-Sayed R., Ibrahim K. (2024): Artificial Intelligence in Veterinary Care: A Review” This review summarizes recent advancements in AI-driven veterinary diagnostics and treatment planning. It covers deep learning, IoT integration, and smart sensing technologies. While informative, the study lacks experimental validation and case-based insights. It provides a broad overview of AI applications in animal health but does not delve into specific implementation strategies for wound detection or infection monitoring, making it more suitable for conceptual understanding. The review emphasizes the growing role of AI in improving diagnostic efficiency and decision-making.

“Li X., Chen H., Brown P. (2022): Deep Learning for Wound Detection and Size Estimation” This paper presents a deep learning framework for automated wound segmentation and size estimation using biomedical imaging. It achieves high accuracy in detecting wound boundaries and measuring affected areas. However, the experiments are conducted on mouse models, limiting applicability to larger animals. Despite this, the study offers valuable insights into scalable wound analysis and supports the development of AI-based veterinary diagnostic tools. The model demonstrates effective feature extraction for precise wound boundary detection.

“Khan S., Verma T., Das P. (2025): Detection of Pet Skin Diseases Using Deep Learning” This work implements CNN models to detect common pet skin conditions such as ringworm, scabies, and ear mite. It explores mobile-based deployment for practical veterinary use and demonstrates promising results. However, the dataset size is limited, which affects the model’s ability to generalize across species and disease types. The study supports real-time diagnostics and highlights the feasibility of AI integration in handheld veterinary tools. The approach also focuses on improving accessibility through mobile platforms.

“Liu Y., Zhao M., Chen K. (2021): AI-Based Pet Wound Classification Using CNNs” This study employs a ResNet-50 CNN architecture to classify pet wound images with high precision. It validates deep learning’s effectiveness in veterinary image analysis and supports automated diagnostics. However, it requires large annotated datasets and high-end GPU resources, which may limit its deployment in low-resource or field environments. The paper is valuable for understanding image- based wound classification in animals.

“Kumar R., Singh A. (2023): Smart Comb with IR Sensor for Wound Detection” This research introduces a smart comb embedded with ESP32 and MLX90614 infrared sensor to detect heat variations on animal skin. It offers a non-invasive method for identifying wounds and supports portable diagnostics. Accuracy declines in animals with thick fur, which limits its effectiveness in certain breeds. The device is suitable for surface-level wound detection and real-time feedback in grooming scenarios.

“Fernandez J., Lee W., Gupta S. (2020): IoT Tools for Infection Monitoring” This paper presents an IoT-based system for real-time infection alerts using platforms like Blynk. It enables remote wound tracking and cloud-based data logging, supporting connected veterinary setups. The system is effective for continuous monitoring but depends on stable internet connectivity, which may restrict its use in rural or offline environments. It highlights the role of IoT in enhancing veterinary diagnostics.

“Rajesh P., Patel N. (2022): YOLOv5-Based Wound Detection in Canine Images” This study applies the YOLOv5 architecture for rapid wound localization in canine skin images. It supports high-speed automated diagnosis and image-based screening. The method is efficient and suitable for real-time applications, but environmental factors like lighting and shadows can cause false positives. Calibration and preprocessing are necessary for consistent results in diverse settings.

“Thomas G., Anil P., Roy K. (2025): Thermal + RGB Fusion for Skin Disease Analysis” This paper integrates thermal imaging with RGB data to enhance skin disease detection accuracy. The multimodal fusion approach improves temperature- based anomaly identification and supports advanced diagnostics. However, it requires precise sensor calibration and synchronization, increasing system complexity. The method is best suited for controlled environments and clinical applications where high accuracy is essential.

“Martinez L., Costa V., Brown P. (2024): IR Thermography for Early Infection Detection” This study explores infrared thermography to detect localized heat patterns indicative of infection and inflammation. It supports early-stage wound monitoring and is effective in clinical and research settings. The technique is non- contact and safe for animals, but the high cost of IR cameras limits accessibility in low- resource environments. It emphasizes the importance of thermal imaging in veterinary care.

“Singh R., Verma D., Kapoor S. (2021): Remote Monitoring of Animal Skin Infections” This work presents a system that streams live temperature and video data for remote diagnosis of animal skin infections. It demonstrates the potential of telemedicine in veterinary practice and supports real-time monitoring. However, short- range wireless transmission limits its deployment in large field environments. The system is ideal for mobile veterinary units and outreach programs.

“Karthik P., Mohan S., Iqbal A. (2022): Lightweight TensorFlow Models for Skin Classification” This study focuses on optimizing TensorFlow models for low-power microcontrollers, enabling AI-driven classification on embedded systems. It supports portable diagnostic tools for veterinary applications and is suitable for field use. Performance depends on curated image datasets, and model accuracy may vary with hardware constraints. The approach is scalable and energy-efficient.

“Reddy B., Sharma H., Lopez A. (2020): Cloud-Integrated Animal Health Diagnostics” This research outlines a cloud-connected diagnostic framework that provides real-time dashboards for tracking animal health metrics. It enables centralized data access and remote decision-making. However, reliance on internet connectivity restricts usability in rural or offline environments. The system is ideal for connected clinics and research labs, supporting scalable veterinary diagnostics.

“Chouhan A., Kumar D., Li M. (2021): ML Algorithms for Veterinary Disease Classification” This study compares multiple machine learning algorithms for classifying various animal diseases. It achieves high accuracy across several datasets and validates AI’s diagnostic capabilities in veterinary science. The method’s dependence on extensive labeled data remains a practical challenge. It supports multi- class disease detection and offers insights into algorithm selection for veterinary applications.

“Prakash K., Noor S., Wang Z. (2023): CNN + IR Sensor-Based Pet Comb for Skin Scanning” This paper combines CNN-based image recognition with infrared sensing in a handheld comb device for wound detection. It enables mobile alerts for veterinary monitoring and supports real-time feedback. However, processing delays occur on devices with limited computing power, affecting responsiveness. The system is suitable for portable diagnostics and grooming-based health checks.

“Sharma R., Mehta P., George L. (2025): UV Diagnostic Device for Ectoparasite Detection” This study develops a UV light-based device for non-invasive detection of ectoparasites such as fleas and ticks in dogs. It enhances diagnostic visibility in early- stage infestations and supports field use. Despite its novelty, the lack of software integration limits automation and data logging. The device is useful for visual screening and grooming applications.

“Varun T., Desai R., Lam K. (2024): Design of Smart Grooming Tools for Veterinary Assistance” This paper presents grooming tools equipped with OLED displays to show skin temperature readings. It supports manual veterinary assessment in field conditions and offers basic diagnostic capabilities. While useful for outreach programs, the system lacks automation and requires active grooming for data collection. It is suitable for low-tech environments and educational use.

“Sharma L., Patel J., Omar Y. (2021): Multimodal AI-Based Veterinary Diagnostic Systems” This study combines visual and thermal data for automated veterinary diagnostics. It supports advanced analysis of skin conditions using AI and enhances detection accuracy through sensor fusion. Integration complexity poses technical challenges, but the system is ideal for clinical settings and research applications. It highlights the future of multimodal diagnostics in veterinary care.

In conclusion, this chapter has reviewed existing literature related to wound detection, IoT-based monitoring, and machine learning applications in veterinary care. It highlighted the strengths of current approaches while identifying key limitations such as lack of real-time wearable solutions, limited sensor integration, and high implementation costs. These findings emphasize the need for a compact, cost-effective, and integrated system capable of providing continuous monitoring and accurate analysis.

The insights gained from this review form the basis for the proposed Smart Glove system, addressing the identified gaps through a combination of image processing, sensor fusion, and cloud-based technologies. Furthermore, the reviewed studies underline the importance of combining multiple data sources to improve diagnostic accuracy. The lack of practical, field-ready solutions highlights the relevance of developing a wearable system for real-time applications. This motivates the integration of hardware and software components in a unified platform. These considerations guide the design and implementation of the proposed system in the following chapters.

The next chapter presents the detailed methodology of the proposed system, including system design, hardware and software components, data processing techniques, and implementation strategy for real-time wound detection and monitoring.

CHAPTER 3

METHODOLOGY

This chapter presents the methodology adopted for the development of the Smart Glove system for real-time wound detection and monitoring in dogs. It describes the overall design approach, including system architecture, hardware components, sensor integration, and software implementation. The chapter also explains the working principle, data processing techniques, and the role of the machine learning model in analyzing wound conditions. The objective is to provide a clear understanding of how the proposed system is designed and implemented to achieve accurate, real-time, and efficient monitoring in a portable and cost-effective manner.

3.1 BLOCK DIAGRAM

The system architecture of the proposed smart glove provides an overview of how the components work together to detect and monitor wounds in dogs. It consists of three main parts: input, processing, and output. The input section includes the ESP32-CAM for capturing wound images and sensors such as temperature, humidity, and colour sensors for collecting relevant data. These inputs are sent to the processing unit, which is the ESP32 DevKit V1.

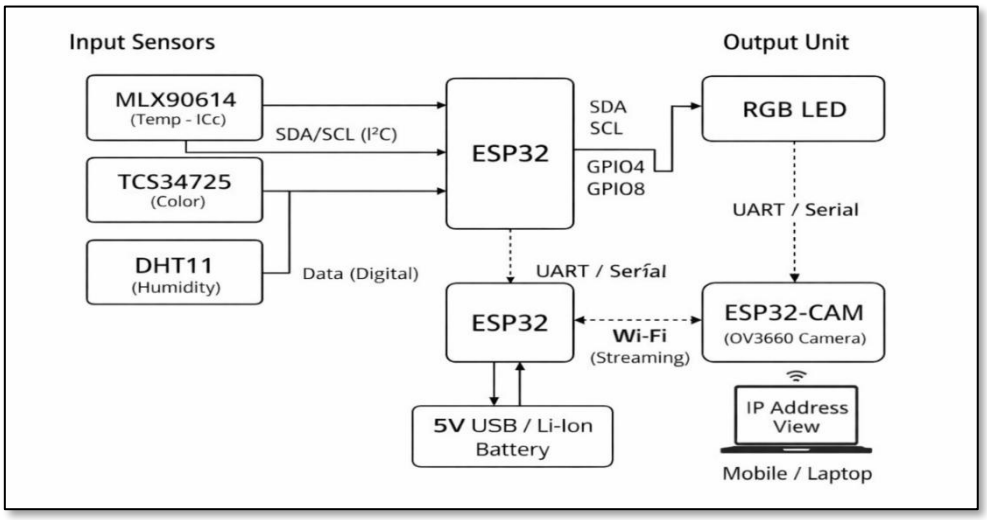

Fig. 3.1 Functional Block Diagram of Smart Glove

In the processing stage, the collected data is preprocessed using techniques like noise removal and thresholding to improve accuracy. The processed image and sensor data are then analyzed using a machine learning model to classify the wound condition. The output section includes the cloud platform ThingSpeak, where the data is stored, and a mobile application developed using MIT App Inventor, which displays the results in real time. An RGB LED is also used to indicate the infection level.

The output section includes the cloud platform ThingSpeak, where the data is stored, and a mobile application developed using MIT App Inventor, which displays the results in real time. An RGB LED is also used to indicate the infection level. This block diagram emphasizes the seamless integration of sensing, processing, and output components in a compact, low-power architecture, making the glove suitable for portable veterinary diagnostic applications.

3.2 HARDWARE SCHEMATIC OVERVIEW

The hardware schematic shows the interconnection of the ESP32 DevKit V1, ESP32-CAM module, sensors, and RGB LED in the proposed system. It explains how sensor data such as temperature, humidity, and color values are collected and processed, while the ESP32-CAM captures wound images for further analysis.

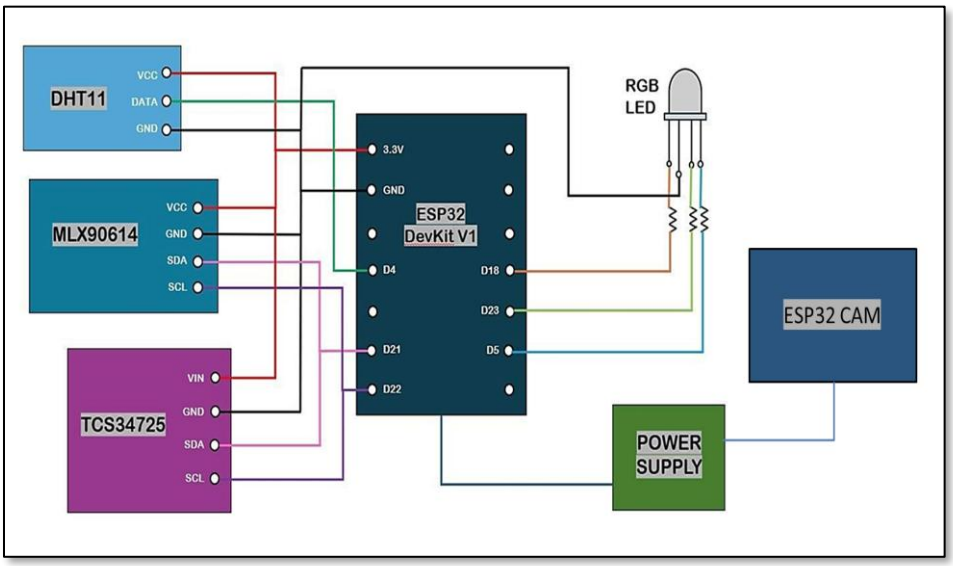

Fig. 3.2 Circuit Diagram

The MLX90614 and TCS34725 sensors are connected using a shared I2C communication bus, while the DHT11 uses a dedicated digital pin. The RGB LED is interfaced with GPIO pins to provide visual indication of the detected condition. The ESP32-CAM operates separately for image capture and communicates through serial programming connections. The design ensures stable operation with proper grounding and regulated power supply, making the system suitable for real-time and wearable applications.

Table 3.1 Sensor and Component Pin Configuration

Component	Pin	Connection	Notes
MLX90614 (IR Temp)	SDA	GPIO 21	Shared I2C Bus
MLX90614 (IR Temp)	SCL	GPIO 22	Shared I2C Bus
TCS34725 (Colour)	SDA	GPIO 21	Shares with MLX90614
TCS34725 (Colour)	SCL	GPIO 22	Shares with MLX90614
DHT11 (Humidity)	Data	GPIO 4	Digital Input
RGB LED	Red	GPIO 2	Current-Limiting Resistor Required
RGB LED	Green	GPIO 16	Current-Limiting Resistor Required
RGB LED	Blue	GPIO 17	Current-Limiting Resistor Required
ESP32-CAM	U0R (RX)	FTDI TX	Serial Programming
ESP32-CAM	U0T (TX)	FTDI RX	Serial Programming
ESP32-CAM	5V	5V Supply	Power Input
ESP32-CAM	GND	GND	Common Ground
ESP32-CAM	GPIO 0	GND (during upload)	Flash Mode
Power	VCC	3.3V / 5V	Shared Power Supply
Power	GND	GND	Common Ground

This configuration enables efficient data acquisition, image capture, and reliable system operation for real-time wound monitoring.

3.3 HARDWARE COMPONENTS

The proposed system consists of several hardware components that work together to perform wound detection and monitoring. The ESP32-CAM module is used to capture images of the wound, while the ESP32 DevKit V1 acts as the main controller for collecting and processing sensor data. An FTDI programmer is used to upload code to the ESP32-CAM module and establish communication during programming. The system also includes an IR temperature sensor to measure wound temperature without contact, a humidity sensor to monitor moisture levels, and a color sensor to detect changes in wound.

In addition to these components, basic electronic elements such as resistors are used for proper circuit functioning, and a perfboard is used to assemble and connect the components securely. A glove is used as the base structure to integrate all the components, making the system wearable and portable. An RGB LED is also included to provide visual indication of the infection level. All these components together form a compact and efficient hardware setup for real-time wound detection and monitoring.

3.3.1 ESP32 CAM

The ESP32-CAM is a highly versatile and cost-effective development board that integrates the ESP32-S system-on-a-chip with an onboard OV3660 camera module, making it suitable for visual IoT applications. This compact module features a dual-core 32-bit LX6 microprocessor operating up to 240 MHz, with 520 KB of internal SRAM and 4 MB of external PSRAM for handling and buffering high-resolution images.

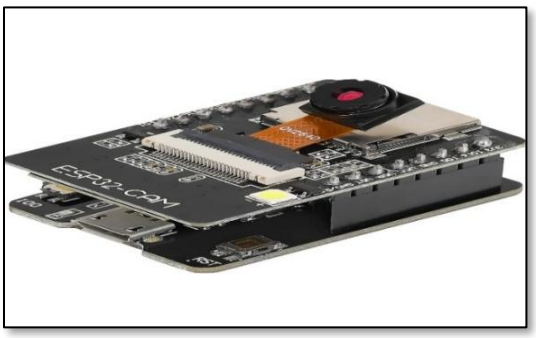

Fig. 3.3 ESP32 CAM

The OV3660 camera module enables clear image capture, supporting various resolutions required for real-time monitoring tasks. The ESP32-CAM also includes 2.4 GHz Wi-Fi and dual-mode Bluetooth (Classic and BLE), allowing efficient wireless data transmission.

Additionally, a microSD card slot is available for local storage, making it useful for saving images and data logging in embedded applications.

3.3.2 ESP32 DevKit V1 (Microcontroller)

The ESP32 microcontroller serves as the brain of the system. It is a low-power, dual- core processor featuring built-in Wi-Fi and Bluetooth. The device supports multiple GPIOs and communication protocols such as I²C and UART, enabling efficient interfacing with multiple sensors.

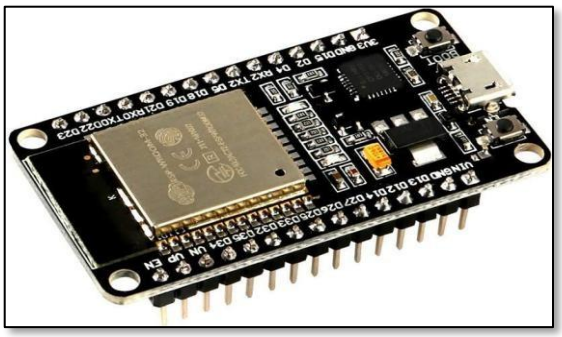

Fig. 3.4 ESP32 DevKit V1

It reads sensor data, executes logical comparisons, and generates the appropriate output signals. Its real-time processing capability ensures timely detection and response to changes in wound conditions.

3.3.3 MLX90614 – Infrared Temperature Sensor

The MLX90614 is a non-contact infrared thermometer used to measure the skin temperature of the wound surface.

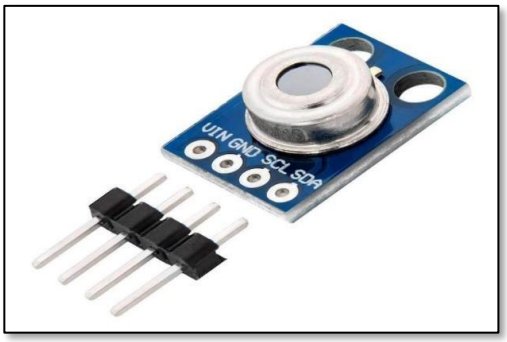

Fig. 3.5 MLX90614 – Infrared Temperature Sensor

It detects emitted infrared radiation and converts it to a digital signal via I²C communication. This allows for hygienic, accurate, and safe measurement. Its wide operating range and high precision make it suitable for detecting subtle temperature variations in infected tissue.

3.3.4 TCS34725 – RGB Colour Sensor

The TCS34725 measures colour intensity and spectral composition in the red, green, and blue channels. It detects changes in wound colour, such as redness or yellowish tones, which correspond to infection or pus formation.

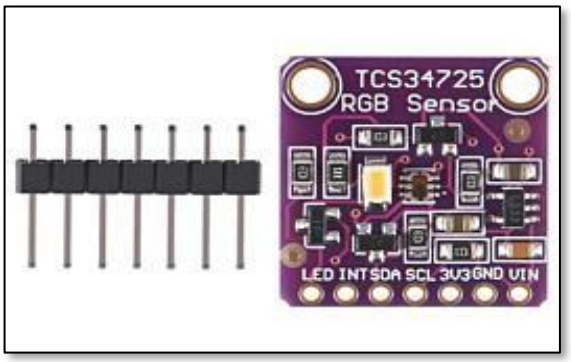

Fig. 3.6 TCS34725 – RGB Colour Sensor

The built-in IR filter improves measurement accuracy by minimizing ambient light interference. It communicates with the ESP32 via the I²C protocol, ensuring efficient data transfer. Its high sensitivity and compact form factor make it ideal for integration into wearable diagnostic systems.

3.3.5 DHT11 – Humidity Sensor

The DHT11 is a low-cost digital sensor used to measure temperature and humidity. It provides calibrated output through a single-wire digital interface, simplifying integration.

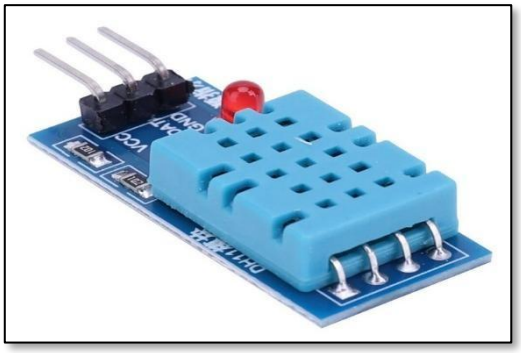

Fig. 3.7 DHT11 – Humidity Sensor

In this project, it monitors ambient conditions around animal wounds for infection detection. Its compact size and low power consumption make it ideal for wearable applications. Consistent readings during testing confirmed its reliability and accuracy in real-time monitoring.

3.3.6 RGB LED

The RGB LED provides a Colour-coded visual indication of the infection status detected by the sensors. It is controlled by the ESP32 GPIO pins, where each Colour represents a condition: green for normal, blue for mild infection, and red for severe infection.

Fig. 3.8 RGB LED

This offers quick and simple visual feedback for real-time monitoring. Its compact form and low power usage make it ideal for wearable diagnostic systems.

3.3.7 Glove Base and FTDI

The sensors are embedded onto a soft, flexible glove material that conforms comfortably to the shape of the user's hand. This glove base provides both flexibility and durability, allowing for natural hand movements while ensuring stable sensor placement.

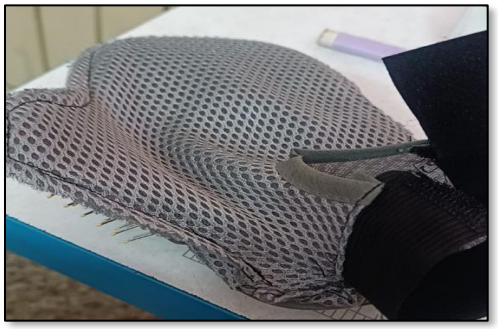

Fig. 3.9 Glove Base

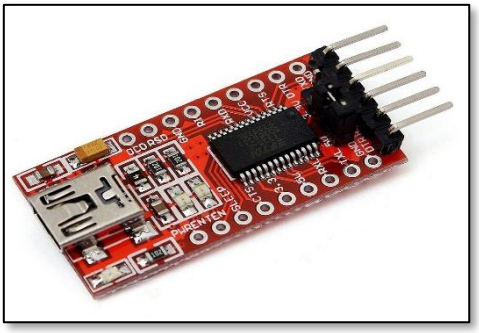

Fig. 3.10 FTDI

The FTDI programmer is used to upload code to the ESP32-CAM module, as it does not have a built-in USB interface. It enables serial communication between the computer and the ESP32-CAM for programming and debugging. The FTDI module converts USB signals to UART signals required by the ESP32-CAM. This ensures reliable data transfer and proper functioning of the camera module.

3.2.8 Power Supply

The system operates on a regulated 5V DC power supply, which can be sourced from a USB port or a portable power bank. This setup ensures consistent voltage levels for all components, including the ESP32 and sensors. The glove is designed for low power consumption, allowing prolonged use in field conditions, while maintaining stable and reliable operation throughout continuous monitoring.

The hardware components of the proposed Smart Glove system are carefully selected and integrated to ensure efficient and accurate wound detection and monitoring. This well-organized hardware setup provides a strong foundation for reliable performance and smooth integration with the software components of the system.

3.4 SOFTWARE TOOLS

The proposed system uses various software tools for programming, data processing, and analysis. Arduino IDE is used to write and upload code to the ESP32-CAM and ESP32 DevKit V1 for controlling image capture and sensor data collection. For image processing tasks such as noise removal and thresholding, Visual Studio Code is used. These preprocessing techniques help in improving the quality of the captured wound images and make them suitable for machine learning analysis. Machine learning algorithms are used to train the model for classifying different types of wound infections based on the processed image and sensor data. The trained model is then used to predict the wound condition. The system also uses the cloud platform ThingSpeak to store processed data, including sensor readings, images, and classification results. A mobile application developed using MIT App Inventor is used to display the results in real time.

3.4.1 Arduino IDE

Arduino IDE is used for writing, compiling, and uploading code to the ESP32 modules. It enables real-time control of sensors, image capture, and system operations.

3.4.2 Google Colab

Google Colab is used for developing and training the machine learning model using wound image datasets. It provides a cloud-based environment with computational support for efficient model training.

3.4.3 Visual Studio Code (VS Code)

VS Code is used for implementing image preprocessing techniques such as noise removal and thresholding. It provides a flexible environment for coding and testing data processing algorithms.

3.4.4 ThingSpeak

ThingSpeak is a cloud platform used to store, analyze, and visualize sensor data and images in real time.

3.4.5 MIT App Inventor

MIT App Inventor is used to develop a mobile application for displaying real-time results. It provides a simple interface for users to view sensor data, images, and classification outputs.

3.5 WORKING PRINCIPLE

The working of the proposed system begins with capturing the wound image using the ESP32-CAM module. At the same time, the ESP32 DevKit V1 collects sensor data such as IR temperature, humidity, and color values from the wound area. The captured image and sensor data are then processed using preprocessing techniques like noise removal and thresholding to improve data quality. After preprocessing, the image and sensor data are given as input to the trained machine learning model. The machine learning model analyzes the data and classifies the wound into different categories based on the infection type.

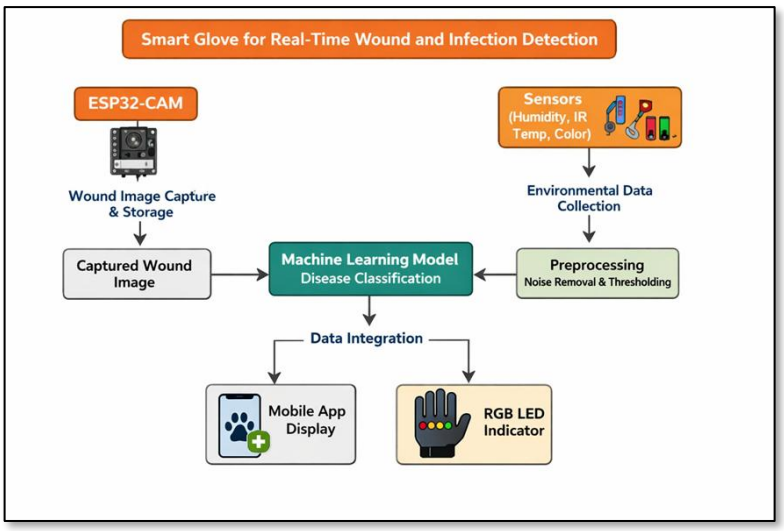

Fig. 3.11 Working Principle

The classification result, along with the processed sensor data and wound image, is then stored in the cloud platform ThingSpeak. A mobile application developed using MIT App Inventor retrieves the data from the cloud and displays it to the user in real time.

3.6 DATA PROCESSING

In the proposed system, the collected data from sensors and the captured wound images are processed to improve accuracy and reliability. The raw image obtained from the ESP32-CAM may contain noise and unwanted variations, which can affect the performance of the machine learning model. To overcome this, preprocessing techniques such as noise removal are applied to enhance the quality of the image. Thresholding is also used to segment the wound area from the background, making it easier to analyze important features. These steps help in extracting meaningful information from the image. Similarly, the sensor data such as temperature, humidity, and color values are also processed to remove errors and fluctuations. This ensures that only accurate and relevant data is used for analysis. The processed image and sensor data are then used as input for the machine learning model, which improves the overall accuracy of wound detection and classification.

3.7 MACHINE LEARNING MODEL

In the proposed system, a machine learning model is used to analyze processed wound images and sensor data for accurate classification of wound conditions. The model is trained using a dataset of four classes: inflammatory infection, healthy, ringworm, and demodicosis. The dataset is divided into training, validation, and testing sets to ensure proper learning and evaluation, with a total of 3001 images.

During training, the model learns important features from the images, while preprocessing techniques such as noise removal and thresholding improve data quality. The trained model is then used to predict the condition of new wound images captured by the system. The model performance is evaluated using accuracy and loss metrics. It achieved a training accuracy of 79.25%, validation accuracy of 80.22% and test accuracy 76%, with a validation loss of 0.9112. A learning rate of 1×10^{-4} is used to ensure stable and efficient training.

3.8 SYSTEM FLOW

The proposed Smart Glove system starts with the ESP32-CAM capturing wound images of the dog. At the same time, the ESP32 DevKit collects sensor data such as humidity, infrared temperature, and color sensor readings. These sensor values are preprocessed using noise removal and thresholding techniques. The processed sensor data and captured images are sent to ThingSpeak via Wi-Fi for real-time cloud storage and monitoring. In parallel, both image and sensor data are given to a machine learning model trained on four wound infection classes to identify the type of wound.

The final prediction is then sent to a mobile application through cloud/API integration, where the user can view the results in real time. An RGB LED is also used in the glove to indicate the detected condition visually. This system enables real-time wound detection, cloud monitoring, and mobile-based output for effective animal health monitoring.

This chapter has presented the detailed methodology of the proposed Smart Glove system, including system architecture, hardware and software components, data processing, and machine learning integration. It explains how different modules work together to achieve accurate and real-time wound detection and monitoring. The structured approach ensures reliability, efficiency, and ease of use in practical applications.

The next chapter focuses on the implementation of the proposed system, detailing the hardware setup, circuit connections, coding, and system integration, and demonstrating how the developed model operates in real-time conditions.

CHAPTER 4

IMPLEMENTATION

This chapter presents the implementation of the proposed Smart Glove system for real-time wound detection and monitoring in dogs. It focuses on the practical realization of the system, including hardware assembly, software development, and integration of all components. The chapter explains how the ESP32-CAM, ESP32 DevKit V1, and various sensors are configured and connected to perform image capture and data acquisition effectively. It also describes the step-by-step working of the system in real-time conditions, including data processing, machine learning-based analysis, and cloud communication. In addition, testing and validation procedures are discussed to ensure the accuracy, reliability, and performance of the system. Overall, this chapter demonstrates how the proposed design is successfully implemented into a functional, efficient, and user-friendly solution for animal healthcare monitoring.

4.1 HARDWARE IMPLEMENTATION

The hardware implementation of the proposed system involves assembling all the components into a compact and wearable glove-based setup. The ESP32-CAM module is used to capture images of the wound, while the ESP32 DevKit V1 is used to collect sensor data such as IR temperature, humidity, and color values.

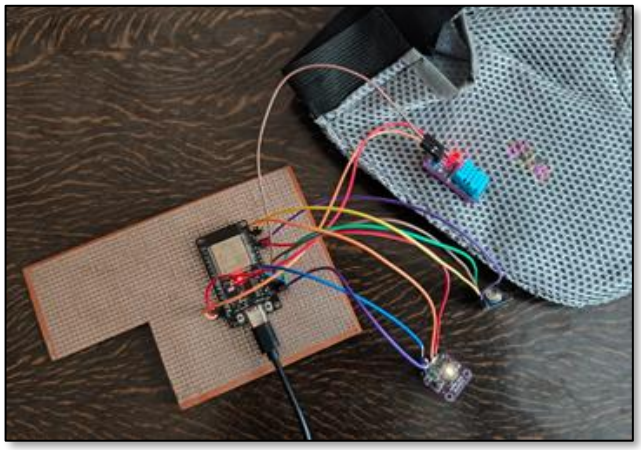

Fig. 4.1 Perfboard Hardware Setup

The components are connected using a perfboard along with resistors to ensure proper electrical connections and stable operation. An FTDI programmer is used to upload code to the ESP32-CAM module. The sensors are positioned in such a way that they can accurately capture data from the wound area. An RGB LED is connected to provide visual indication of the infection level based on the output. All the components are mounted on a glove, making the system portable, user-friendly, and suitable for real-time wound monitoring applications.

4.2 SOFTWARE IMPLEMENTATION

The software implementation of the proposed system focuses on programming, data processing, and result visualization. The ESP32-CAM and ESP32 DevKit V1 are programmed using Arduino IDE to perform image capture and sensor data acquisition. The captured wound images are processed using Visual Studio Code, where techniques such as noise removal and thresholding are applied to improve image quality. Sensor data is also cleaned and prepared for analysis to ensure accuracy.

Fig. 4.2 ESP32-CAM Initialization Output

A machine learning model is trained using the prepared dataset to classify different wound conditions.

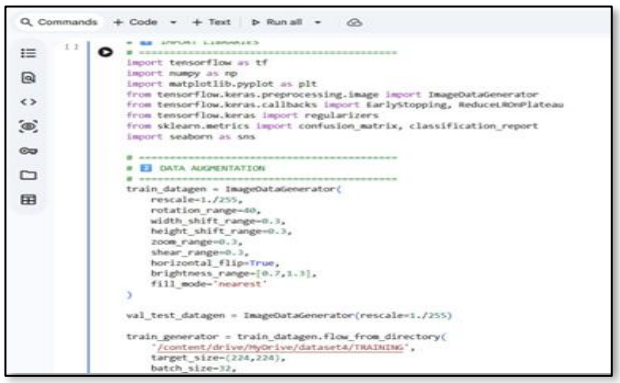

Fig. 4.3 Machine Learning Training Code

The trained model is then used to analyze new input data and generate predictions. The processed data, including sensor readings, wound images, and classification results, are uploaded to the cloud platform ThingSpeak. A mobile application developed using MIT App Inventor retrieves this data and displays it in real time.

4.3 SYSTEM INTEGRATION

The system integration phase combines all the hardware and software components to function as a complete and coordinated system. The ESP32-CAM captures the wound image, while the ESP32 DevKit V1 simultaneously collects sensor data such as temperature, humidity, and color values from the wound area. The collected image and sensor data are processed and prepared for analysis. The processed data is then given to the machine learning model, which classifies the wound condition based on the learned patterns. This ensures accurate and reliable detection of different types of infections.

The classification results, along with the processed data, are uploaded to the cloud platform ThingSpeak. The mobile application developed using MIT App Inventor retrieves this data and displays it to the user in real time. At the same time, the RGB LED provides immediate visual feedback based on the detected condition. This integration ensures smooth communication between all modules and enables efficient real-time monitoring of the system.

4.4 WORKING IMPLEMENTATION

The working implementation of the system begins when the smart glove is placed near the wound area. The ESP32-CAM captures the image of the wound, while the ESP32 DevKit V1 simultaneously collects sensor data such as IR temperature, humidity, and color values.

The captured image and sensor data are then processed using preprocessing techniques like noise removal and thresholding to improve accuracy. The processed data is given as input to the trained machine learning model, which analyzes the wound condition and classifies it into different categories.

The classification result, along with the processed sensor data and wound image, is uploaded to the cloud platform ThingSpeak. A mobile application developed using MIT App Inventor retrieves this data and displays it in real time. Based on the output, the RGB LED provides immediate visual indication of the infection level.

4.5 TESTING AND VALIDATION

The proposed system is tested to ensure proper functioning of both hardware and software components. Initially, individual components such as the ESP32-CAM, sensors, and RGB LED are tested separately to verify their performance and accuracy.

After individual testing, the complete system is tested by capturing wound images and collecting sensor data simultaneously. The collected data is processed and given to the machine learning model to check the correctness of classification results. The output displayed in the mobile application is verified for accuracy and consistency.The system is also validated using the test dataset to evaluate the performance of the machine learning model. Metrics such as accuracy and loss are used to measure the effectiveness of the model.

4.6 CHALLENGES AND LIMITATIONS

During the implementation of the proposed system, several challenges were encountered. One of the main challenges is capturing clear wound images using the ESP32-CAM, as lighting conditions and camera quality can affect image clarity. Variations in sensor readings due to environmental factors may also impact the accuracy of the system. Another limitation is the dependency on the trained dataset. If the dataset does not cover all possible wound conditions, the machine learning model may not provide accurate classification for unknown cases. Additionally, slight variations in wound appearance may lead to misclassification. The system also depends on proper placement of sensors on the glove to obtain accurate readings. Any misalignment can affect the data quality. Furthermore, real-time performance may be affected by processing time and cloud communication delays.

This chapter has detailed the implementation of the proposed Smart Glove system, including hardware setup, software development, and system integration. It demonstrates how all components work together to achieve real-time wound detection and monitoring, ensuring proper functionality, reliability, and ease of use in practical applications.

The next chapter presents the results and discussion of the system, analyzing the performance of hardware and software components, along with the accuracy of the machine learning model and overall system effectiveness.

CHAPTER 5

RESULTS AND DISCUSSIONS

This chapter presents the results and discussion of the proposed Smart Glove system developed for real-time wound detection and monitoring in dogs. It evaluates the performance of both hardware and software components, along with the effectiveness of the machine learning model used for classification. The chapter highlights how the system performs under real-time conditions, including data acquisition, processing, and visualization. It also analyzes the accuracy and reliability of the model through training results and confusion matrix evaluation. Furthermore, the integration of all system components is discussed to demonstrate the overall functionality and efficiency of the proposed system. The observations and findings provide insights into the strengths and limitations of the system, helping to assess its practical applicability in animal healthcare.

5.1 PROTOTYPE MODEL

The developed smart glove prototype was successfully assembled, resulting in a compact, lightweight, and easy-to-use design. All components are integrated onto the glove, ensuring flexibility and stable operation during use.

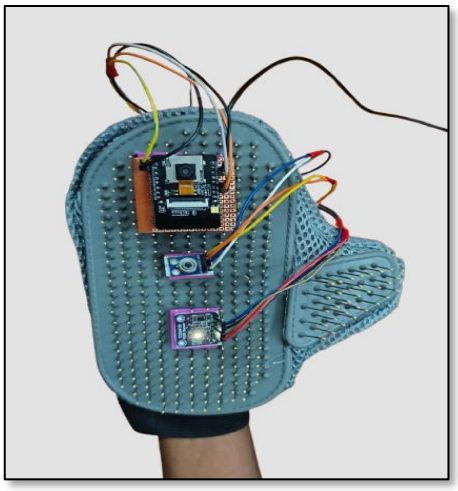

Fig. 5.1 Front View of the Prototype

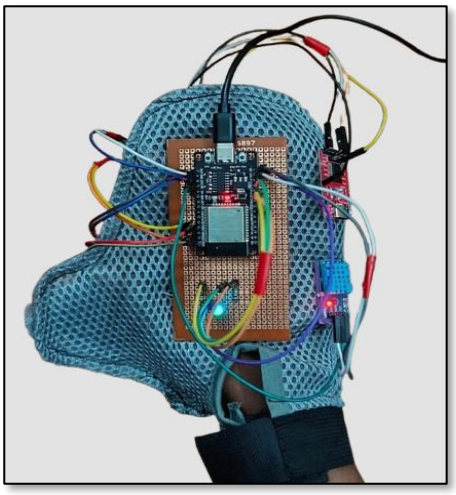

Fig. 5.2 Back View of the Prototype

The front view shows the placement of key components such as the TCS34725 color sensor, MLX90614 infrared temperature sensor, and ESP32-CAM module, which are positioned to capture wound images and sensor data in real time. The back view shows the ESP32 DevKit V1 and DHT11 humidity sensor mounted securely along with the supporting circuit, ensuring stable functioning and easy handling. Overall, the prototype achieves a balance between functionality, portability, and user convenience for veterinary applications.

5.2 HARDWARE RESULTS

The hardware components of the proposed smart glove system were tested to evaluate their performance and reliability. The ESP32-CAM module was able to capture wound images with acceptable clarity under proper lighting conditions.

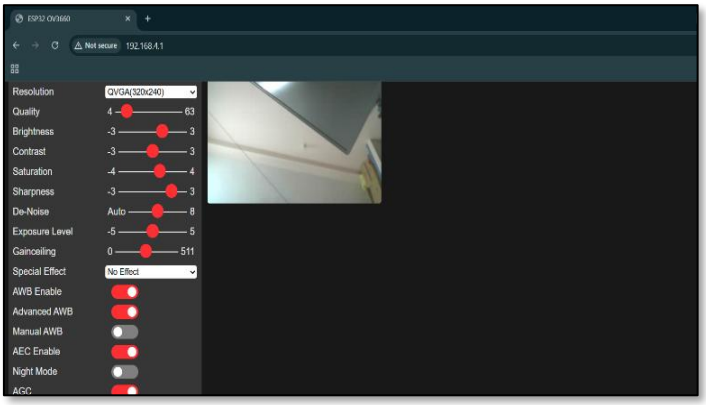

Fig. 5.3 ESP32-CAM Output through IP Address

Arduino IDE was utilized for programming the system and monitoring sensor outputs in real time, enabling efficient data acquisition and smooth integration of all components. The ESP32 DevKit V1 effectively collected sensor data such as IR temperature, humidity, and color values with minimal delay.

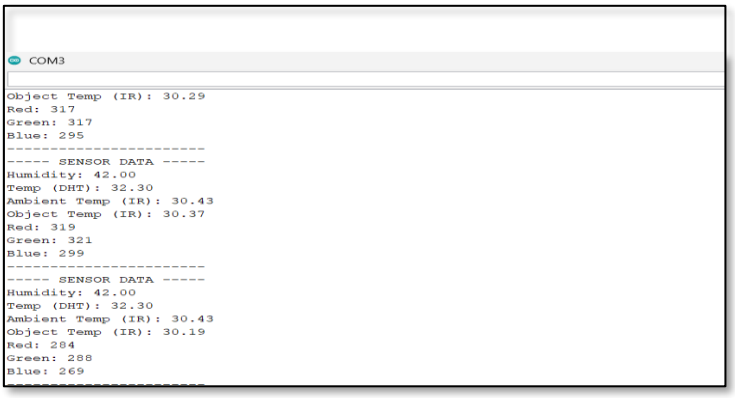

Fig. 5.4 Serial Monitor Output Showing Sensor Readings

The IR temperature sensor provided consistent non-contact readings, while the humidity and color sensors accurately detected moisture levels and variations in wound color. The RGB LED responded correctly by indicating different infection levels based on the output. All components were assembled using a perfboard, ensuring stable circuit operation and ease of handling. The glove-based design further enhanced portability and usability. Overall, the hardware system demonstrated reliable performance for real-time wound monitoring applications.

5.3 SOFTWARE RESULTS

The software components of the proposed system were tested to evaluate their functionality and performance. The ESP32 modules were successfully programmed using Arduino IDE for image capture and sensor data acquisition. The captured wound images were processed using Visual Studio Code, where preprocessing techniques such as noise removal and thresholding improved image quality.

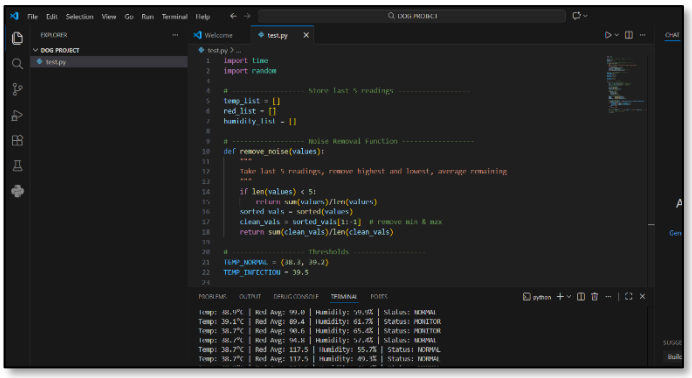

Fig. 5.5 VS Code – Noise Removal and Thresholding Output

The processed data, including wound images and sensor readings, were successfully uploaded to the cloud platform ThingSpeak without major delay.

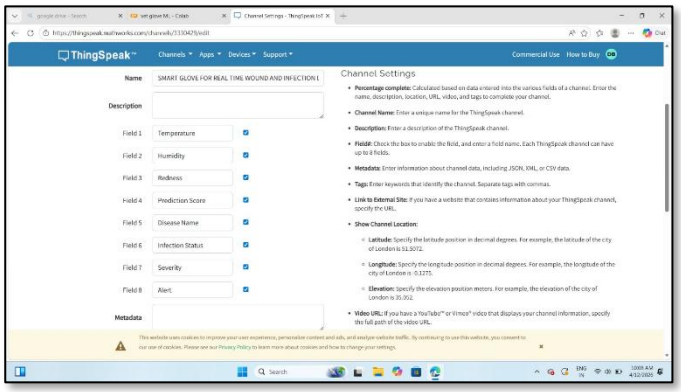

Fig. 5.6 ThingSpeak Cloud Output Screen

The mobile application developed using MIT App Inventor was able to retrieve and display the data in real time.

Fig. 5.7 Mobile Application Display

The application interface clearly showed the wound image, sensor values, and classification results, making it easy for users to understand the condition. Overall, the software system performed efficiently and ensured smooth data processing, storage, and visualization.

5.3.1 MODEL TRAINING AND VALIDATION RESULTS

The machine learning model was trained using a dataset of 3001 images consisting of four classes: inflammatory infection, healthy, ringworm, and demodicosis. The dataset was divided into training, validation, and testing sets to ensure proper evaluation of the model.

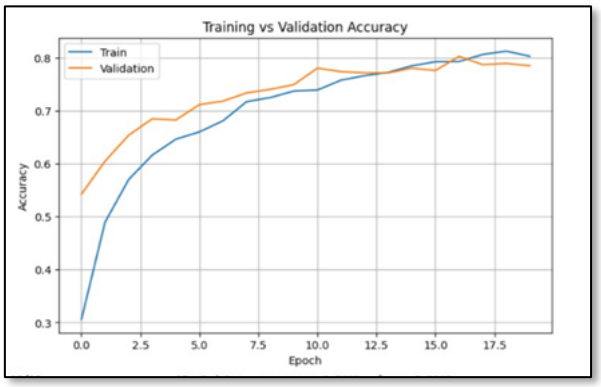

Fig. 5.8 Training and validation accuracy graph

During training, the model was able to learn important features from the wound images, which helped in classifying different types of infections. The model achieved a training accuracy of 79.25%, a validation accuracy of 80.22% and test accuracy of 76%, indicating consistent performance and good generalization on unseen data. The validation loss obtained was 0.9113, which represents the difference between predicted and actual outputs. The learning rate used during training was 1×10^{-4}, ensuring stable and gradual learning of the model.

Fig. 5.9 Model Prediction Output

Shows the output of the trained machine learning model, including prediction results and accuracy metrics. Displays the classified wound image, demonstrating the model's ability to identify infection types.

5.3.2 CONFUSION MATRIX ANALYSIS

The confusion matrix is used to evaluate the performance of the machine learning model by comparing the actual and predicted classifications of wound conditions. It provides a clear understanding of how well the model identifies each class, including inflammatory infection, healthy, ringworm, and demodicosis.

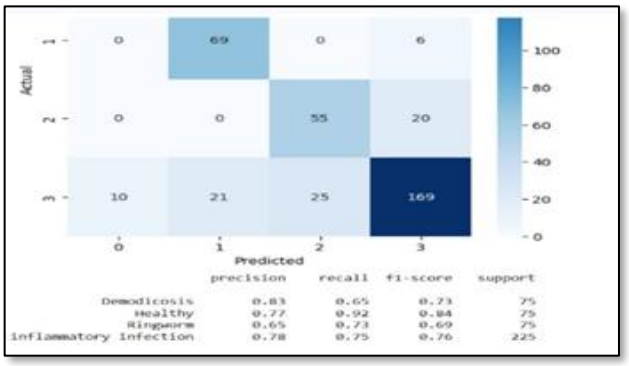

Fig. 5.10 Confusion matrix for multi-class classification

The results show that the model correctly classifies most of the samples in each category, indicating good performance. Higher values along the diagonal of the confusion matrix represent correct predictions, while lower values in other positions indicate fewer misclassifications.

Some misclassification is observed between classes with similar visual features, such as inflammatory infection and ringworm. This is mainly due to overlapping characteristics in wound appearance. However, the overall classification accuracy remains satisfactory.

5.4 SYSTEM INTEGRATION RESULTS

The complete system was tested by integrating all hardware and software components to evaluate real-time performance. The ESP32-CAM successfully captured wound images while the ESP32 DevKit V1 simultaneously collected sensor data such as IR temperature, humidity, and color values.

The collected data was processed and analyzed using the trained machine learning model, which accurately classified the wound condition. The results, along with sensor readings and images, were successfully uploaded to the cloud platform ThingSpeak. The mobile application developed using MIT App Inventor retrieved and displayed the data in real time without noticeable delay. The RGB LED also provided immediate visual indication based on the classification result.

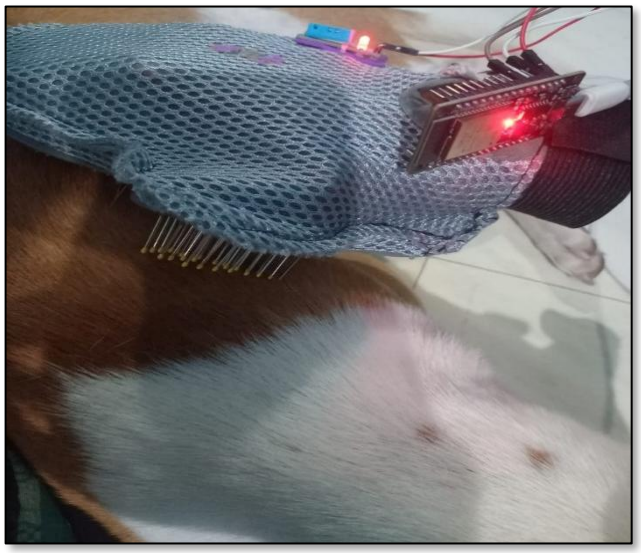

Fig. 5.11 Real-Time Testing of Smart Glove on Dog

5.5 COMPARISON WITH EXISTING SYSTEM

The developed smart glove was compared with conventional wound detection and diagnostic methods. Its low-cost, user-friendly design makes it particularly beneficial for rural or resource-limited veterinary centers.

Table 5.1 Comparison Between Existing system and Proposed System

Parameter	**Convolutional Neural Networks for Wound Detection**	**Smart Glove for Real-Time Wound and Infection Detection**
Approach	Pure Computer Vision (CNN)	Multimodal Sensor Fusion + AI
Input	RGB Image Data	Visual (OV3660) + Thermal + Humidity + Color
Training Accuracy	71.20%	79.25%
Validation Accuracy	68.45%	80.22%
Validation Loss	1.1500	0.9113
Test Accuracy	~ 69.00%	~ 76.00%
IoT Support	None	Cloud + Mobile App
Portability	Non-Portable	Wearable Glove
Hardware Integration	PC-based	Dual-Node (ESP32 DevKit V1 & ESP32-CAM)

The comparison indicates that the proposed system achieves better performance in terms of accuracy and loss by incorporating multimodal sensor data along with image analysis. Unlike conventional CNN-based approaches, it supports real-time monitoring with integrated hardware and IoT capabilities. The wearable design further enhances portability, making it more suitable for practical veterinary applications.

5.6 DISCUSSION

The results obtained from the proposed system show that the integration of hardware, software, and machine learning provides an effective solution for real-time wound detection and monitoring in animals. The hardware components performed reliably, and the software system ensured smooth data processing and visualization. The machine learning model achieved satisfactory accuracy and was able to classify different wound conditions effectively.

The confusion matrix analysis further confirmed that most of the predictions were correct, with only minor misclassifications between similar classes. The system integration results demonstrated that all components worked together without major issues, and real-time data transmission and display were achieved successfully through ThingSpeak and MIT App Inventor. However, the system performance may be affected by factors such as lighting conditions, sensor placement, and dataset quality. Improving these factors and increasing the dataset size can further enhance the accuracy and reliability of the system.

This chapter has presented the results and discussion of the proposed Smart Glove system, highlighting the performance of hardware, software, and the machine learning model. The system demonstrated reliable data acquisition, effective image processing, and satisfactory classification accuracy during real-time testing. The integration of all components enabled smooth operation and user-friendly monitoring, confirming the effectiveness of the proposed approach for animal wound detection.

The next chapter summarizes the overall outcomes of the project and highlights possible improvements and enhancements that can be implemented to further increase the efficiency, accuracy, and usability of the system.

CHAPTER 6

CONCLUSION

The proposed smart wearable system for real-time wound and infection detection in dogs has been successfully designed, implemented, and evaluated. This system combines embedded hardware, sensor technology, and machine learning techniques to provide an efficient and intelligent solution for veterinary wound monitoring. The integration of ESP32 DevKit V1, ESP32-CAM, and multiple sensors such as the MLX90614 infrared temperature sensor, DHT11 humidity sensor, and TCS34725 color sensor enables the system to collect both physiological and visual data from the wound area. The use of a multi-sensor approach enhances the reliability of the system by capturing different aspects of the wound condition, including temperature variation, moisture levels, and color changes. These parameters are crucial indicators of infection and inflammation. The ESP32-CAM module captures real-time images, which are analyzed using a deep learning model based on the MobileNet architecture. The use of MobileNet ensures efficient computation and suitability for real-time applications with limited hardware resources.

To improve model performance, techniques such as early stopping and fine-tuning were applied during training. Early stopping helps to prevent overfitting by stopping the training process when validation performance does not improve, while fine-tuning enhances the model's ability to learn domain-specific features. However, the dataset used for training contained images with high similarity and some blurred samples, which made the classification task more challenging, especially for visually similar skin conditions.

In addition to image-based analysis, sensor data is processed using preprocessing techniques such as noise removal and threshold filtering. These techniques improve data quality and reduce environmental disturbances, thereby enhancing system performance. The hybrid combination of sensor data and image-based classification increases the robustness and reliability of the system compared to traditional single-parameter approaches. The performance of the system was evaluated using training, validation, and test datasets. The model achieved a training accuracy in the range of 79.25%, validation accuracy of 80.22%,

and test accuracy of 76%. These results indicate consistent and stable performance across different datasets, demonstrating good generalization capability. Although the accuracy is moderate compared to large-scale deep learning models, it is suitable for real-time practical applications and prototype-level deployment. The system also incorporates IoT capabilities, enabling real-time monitoring through a mobile application and cloud platform. This allows veterinarians and pet owners to access diagnostic results remotely, improving convenience and decision-making. The wearable glove design further enhances usability by ensuring proper sensor placement and ease of operation in both clinical and field environments.

Overall, the proposed system provides a non-invasive, portable, and cost-effective solution for early detection of wound infections in dogs. By integrating hardware, software, and intelligent analysis, the system serves as a reliable screening tool that supports early diagnosis and timely treatment, ultimately improving animal health and welfare.

6.1 FUTURE SCOPE

The proposed system provides a strong foundation for further advancements in smart veterinary healthcare. One of the primary areas for improvement is the expansion and enhancement of the dataset used for training the deep learning model. The current dataset contains images with high similarity and some blurred samples, which affect classification performance. In future work, collecting a larger and more diverse dataset with improved image quality, different lighting conditions, and varied disease stages can significantly enhance model accuracy and generalization.

The current system utilizes the MobileNet architecture, which is efficient for real-time applications. In this work, techniques such as fine-tuning and data augmentation (including flipping, rotation, and noise addition) have already been applied to improve model performance. However, further improvements can be achieved by optimizing hyperparameters and experimenting with advanced architectures such as ResNet, EfficientNet, or MobileNetV3 to achieve higher classification accuracy, especially for visually similar skin conditions. The system can be further enhanced by integrating additional sensors such as pH sensors, gas sensors, or biosensors to provide more comprehensive wound analysis. This multi-modal approach can improve the detection of infection severity and provide deeper insights into wound conditions.

Another important direction for future work is the implementation of edge computing. Deploying the optimized MobileNet model directly on the ESP32 or an edge device can reduce dependency on cloud platforms, minimize latency, and enable faster real-time predictions even in low-connectivity environments. The mobile application can also be upgraded with advanced features such as real-time alerts, push notifications, historical data tracking, graphical visualization of sensor readings, and automated report generation. Integration with cloud-based IoT platforms can allow centralized monitoring and remote access for veterinarians.

Furthermore, the system can be extended to support multiple animal species, including cats and livestock, increasing its applicability in broader veterinary and agricultural domains. Improvements in hardware design can also make the wearable device more compact, energy-efficient, and user-friendly. Overall, with continuous improvements in dataset quality, model optimization, sensor integration, and IoT technologies, the proposed system has the potential to evolve into a highly accurate, intelligent, and scalable solution for real-time wound monitoring and disease detection in veterinary healthcare.

CHAPTER 7

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